Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/20/2012
PI Name:	Naidu, Mamta Ph.D.		
Project Title:	High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mamta.naidu@griresearch.org	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	(617) 444-3050
Organization Name:	GeneSys Research Institute/Tufts University School of Medicine		
PI Address 1:	Center for Cancer Systems Biology/Department of Medicine		
PI Address 2:	736 Cambridge Street, CBR 3		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02135	**Congressional District:**	7
Comments:	NOTE: Previous affiliation was State University of New York-Stony Brook, from October 2011-December 2012. Prior to that, affiliation was Brookhaven National Laboratory, Upton, NY, until September 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	09/15/2011	**End Date:**	06/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2013 (from 12/13/2013) due to PI move to new institution, per A. Chu/ARC (Ed., 1/29/2013) NOTE: Extended to 12/13/2013, per NSSC information (Ed., 12/19/12)		
Key Personnel Changes/Previous PI:	NOTE: New PI is Bruce Demple, per PI Naidu's move to GenSys Research Institute in January 2013 (Ed., 1/29/13) New Post Doctoral Fellow on this grant since April 2012, owing to move between institutions and SUNYSB procedures for transfer of H1; took 4 months to come through.		
COI Name (Institution):	Demple, Bruce (SUNY-Stony Brook/ Adminstrative PI)		
Grant/Contract No.:	NNX11AO89G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note: Grant is a continuation at SUNY-Stony Brook per PI's move in September 2011 from Brookhaven National Laboratory. See task, "High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo" (NNA10DA32I) for previous reports.] NASA's radiation research program is focused on assuring that humans can travel in space without exceeding the limiting risks from radiation, and on advancing the understanding of response mechanisms to radiation-caused DNA damage. Radiation studies on the CNS have predominantly focused on neurons, with few studies directly addressing the role of glial cells. The focus of our research proposal will be to determine the major DNA repair pathway(s) induced by HZE radiation in glial cells. Oligodendrocytes (OL) are responsible for forming and maintaining myelin, the insulating sheaths of neuronal axons. The damage to OL causes neurological impairment as OL have a slow turnover and are repopulated by resident OL progenitor cells (OPC). Astronauts may encounter repeated exposure to charged particles over the course of their missions such as traveling to Mars and this exposure to low doses of cosmic rays has the potential to prevent astronauts from successfully meeting mission requirements. Base excision repair(BER) is one of the predominant DNA damage repair pathways up regulated in response to oxidative stress by low LET radiation. Also, Apurinic endonuclease 1(Ape1), a key BER player may be involved in adaptive response by linking repair to oxidative pathways, thus, it becomes important to study its role in fractionated versus single low dose of high LET radiation. As Ape1 is known to affect the G1/S transition in hematopoietic embryoid bodies (EB) cells, it also becomes important to determine if cell cycle dependent changes in Ape1 levels can be correlated to differentiation capacity of CG-4. Thus, our project will determine the effects of high LET radiation on the nuclear and mitochondrial BER in two systems; the central glia-4 (CG-4), a glial progenitor cell line that can differentiate into OL or astrocytes, and in OPC cells isolated from rat spinal cord. The latent CNS effects are due to altered DNA repair resulting in altered OPC differentiation causing neuro-inflammation and decreased cognitive function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study of the effects of fractionated, low dose versus single low dose of charged particles on glial cells of brain would help us understand the risk to astronauts exposed to chronic radiation exposures and the risk of normal tissues of patients receiving fractionated doses to their brain tumors. Our study will answer a critical question, "Would lower doses of higher Z particles produce more subtle latent changes, for example, changes in CNS cell population type that do not produce externally obvious deficits until months or years after exposure?" The oxidative stress produced by particles affects DNA repair capacity and our studies will show which DNA repair pathways are affected and how they affect the cells' differentiation responses to stress, thereby possibly altering cognitive behaviour, finally leading to better understanding of risks out there for our astronauts on long term missions.
Task Progress:	From our past 3 year studies, we have been able to clearly show that BER protein, APE1 is upregulated with increased localization to mitochondria with fractionation, predominantly due to inhibition of mitochondrial membrane potential which may result is skewing of differentiation of glial progenitor cells (in vitro) into immature oligodendrocytes and astrocytes, with lower formation of mature oligodendrocytes, which are the myelin forming cells of CNS. In vivo studies in rat spinal cord have shown destruction of both white and gray matter, with ongoing studies to confirm this using cell markers. In addition, we have also been able to show a clear cognitive defect earlier in Si exposed rats (but not Fe exposed rats) and latent (9 months post exposure) proton exposed rats as compared to X-rays. Also, our most recent NSRL studies reveal huge debilitative tumor induction effect in 3/15 Si exposed rats by 4 months of exposure with 0.5 Gy dose, with tumor formation in brain.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Naidu M, Chaudhary P, Sanchez Z. "IA-6, an indazole analog of Hycanthone, a more efficient direct inhibitor of Apurinic Endonuclease 1 (APE1), than Lucanthone/ Hycanthone, is also less toxic." Poster presented at 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. p. 118. , Aug-2011
Abstracts for Journals and Proceedings	Chaudhary P, Sanchez Z, Naidu M. "Fractionated ionizing radiation skews differentiation of glial / oligodendrocyte progenitor cells, with lower mature oligodendrocytes formation, as compared to single low dose of X-rays and particle irradiation." Invited talk. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7073.pdf , Sep-2011
Abstracts for Journals and Proceedings	Kumar MA, Chaudhary P, Shawan J, Biegon A, Naidu M. "Fractionated ionizing radiation skews differentiation of glial/oligodendrocyte progenitor cells and induces cognitive defects." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012